AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 4 · STUDENT LESSON

Methods results and limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

State what was done. Report what was found. Bound what it means.

Separate methods, results, limitations and proposed mechanisms with reasons and legitimate overlap.

Keep one habit: Role contract → twenty-query classification → five-photo picture → interpretation probes → passage rewrite → independent reviewer/cutoff case.

MAIN LESSON · SECTION 1

Keep procedure, finding and interpretation limits connected

> Classify each statement by its role without forcing every sentence into only one box.

Learning goals

I can explain the **mechanism** and terminology of **method**s **result**s and **limitation**s.

I can apply the stated procedure to a new case with a traceable **result**.

I can identify an assumption or **limitation** and design a revealing follow-up.

Retrieve claim scope and inspected original evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Finding primary research. Explain why a recorded **result** needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **method**s **result**s and **limitation**s using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **method**s **result**s and **limitation**s

Apply the stated procedure to a new case with a traceable **result**

Identify an assumption or **limitation** and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define the role each statement plays in the argument

> A limitation bounds a result rather than erasing the reported observation.

Begin with the three-sentence fictional report

Imagine a report says “We tested twenty fictional questions; sixteen passed; no real students used the system.” These statements answer different questions: what was done, what was observed and what the observations cannot settle. Keep them linked. A method detail may also explain a limitation.

A proposed reason for the result is an explanation to investigate, not another observation simply because it sounds plausible.

Define methods, results and limitations

Methods state what was done; **result**s state observed outcomes; **limitation**s state restrictions on interpretation. Mixing them hides missing **evidence**.

Complete passage and classification contract

Statement role	Complete classroom distinction
Method	What was done: cases, versions, scoring, conditions and data collection
Result	What the supplied record reports, with its metric and denominator

Statement role	Complete classroom distinction
Limitation	What that procedure/evidence does not establish or what uncertainty remains
Proposed explanation	A candidate mechanism to test; not an additional observed result
Main supplied passage	Used 20 fictional queries; 16 passed; real student use not studied
Main arithmetic	16/20 = 80% on those fictional queries
Missing details	Full pass rubric, case records and sampling protocol are not supplied by the short passage
Separate picture	Two frozen classifiers, same five labelled leaf photos, one camera, no outdoor photos
Picture result	A 3/5 = 60%; B 4/5 = 80%, difference 20 percentage points
Scope	Fictional reported evidence, not a new model experiment, classroom-learning study or outdoor deployment evaluation

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the role and support of each sentence

> Tie the result to its procedure and denominator before interpreting it.

Original statement-role procedure

Classify each statement by role and link **result**s to the procedure and denominator. A proposed **mechanism** is an explanation to test, not an extra **observation**.

Complete main classification and faithful rewrite ledger

Main supplied statement	Role and reason	Faithful use
Used twenty fictional queries	Method and scope: defines what was evaluated	Describe those query cases, not real learners
Sixteen passed	Reported result; needs total and pass rule	16/20 = 80% on the supplied cases
Real student use was not studied	Limitation and study-scope detail	No classroom effectiveness result supplied
“It worked because students learned faster”	Unsupported proposed explanation/outcome	Neither student use nor learning speed was measured
Full rubric and raw case identities absent	Evidence gap in the short summary	Ask for them; do not invent reproducibility details

State the missing method details

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the **result** in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Methods results and limitations — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Statements: used 20 fictional queries; 16 passed; real student use was not studied.	Classify each statement by role and link results to the procedure and denominator. A proposed mechanism is an explanation to test, not an extra observation.	First is method/scope; second result 16/20 = 80%; third limitation. No classroom effectiveness result is supplied.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict an outdoor-farming conclusion from indoor-only evidence

> A broader application needs evidence that the supplied study does not contain.

Current task and response

- Statements: used 20 fictional queries; 16 passed; real student use was not studied. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the stronger outdoor claim

- Change: The conclusion says B is best for outdoor farming.

▼ **Check after predicting**

Prediction: Reject that extension without outdoor **evidence**.

Reason: The reported conditions do not include the proposed use population.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the leaf-photo method, observed count and missing outdoor evidence

> The correct local finding stays valid when its scope is stated.

Main twenty-query passage and checked roles

Statements: used 20 fictional queries; 16 passed; real student use was not studied.

First is **method**/scope; second **result** 16/20 = **80%**; third **limitation**. No classroom effectiveness **result** is supplied.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate five-photo inputs and full role classification

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.

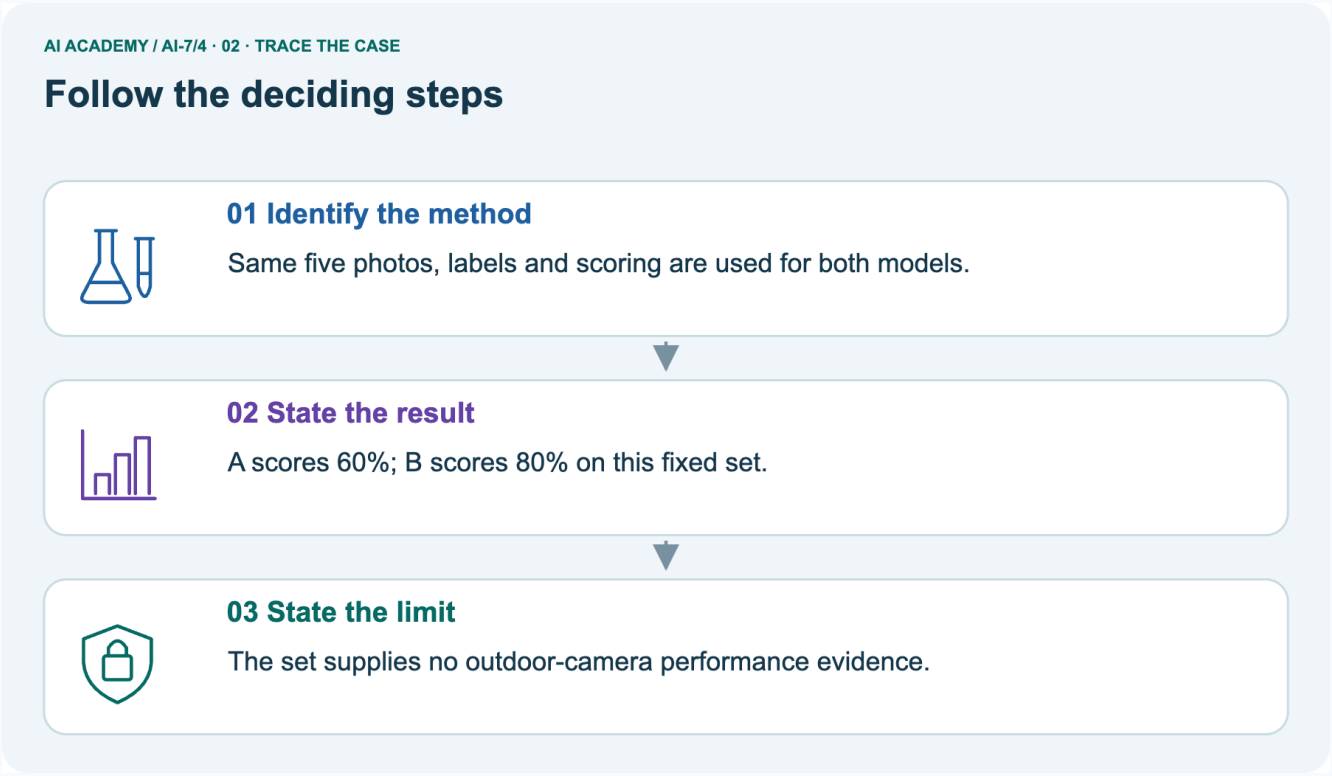
A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

The picture is a separate five-photo evaluation. Method: two frozen classifiers use the same five labelled leaf photos and scoring, from one camera, with no outdoor photos. Result: A is correct on 3/5 = 60%, B on 4/5 = 80%, so B is higher by 20 percentage points on this fixed set. Limitation: outdoor conditions, other cameras and broader sampling are not evaluated.

“B is best for outdoor farming” goes beyond the available comparison. The five-photo result neither proves that B will generalize nor proves it will fail outdoors. It also does not identify an internal causal mechanism.

Separate picture statement	Role	Evidence-faithful wording
Two frozen classifiers; same five labelled photos, one camera	Method	State versions, same photos, labels and scoring conditions
A 3/5 and B 4/5	Reported result	A 60%, B 80%; B higher on these five cases
No outdoor photos	Method/scope detail and limitation	Outdoor performance unassessed
B best for all outdoor farming	Unsupported conclusion	Replace with the fixed-set comparison and explicit missing outdoor evidence

Trace procedure, finding and scope limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify the method	Same five photos, labels and scoring are used for both models.
2. State the result	A scores 60%; B scores 80% on this fixed set.
3. State the limit	The set supplies no outdoor-camera performance evidence.

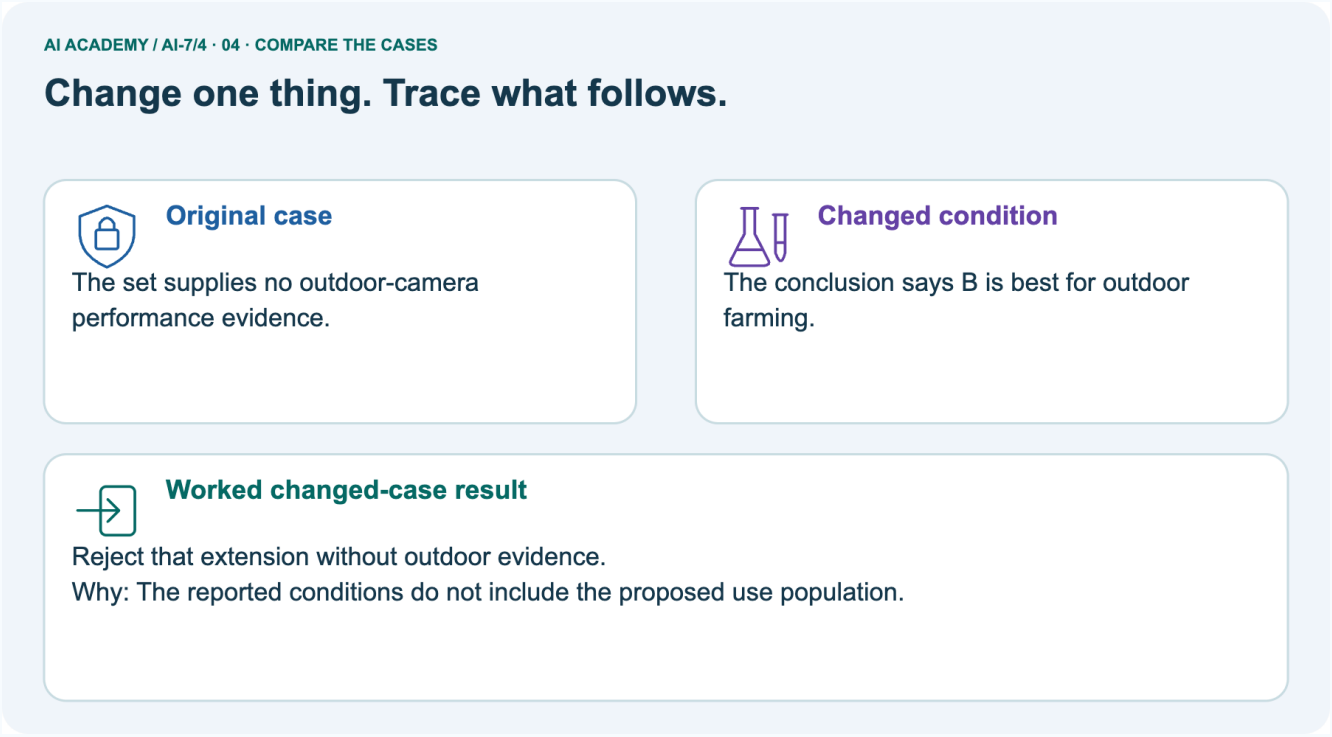
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Methods tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

Predict “best for outdoor farming”

- Change one thing: The conclusion says B is best for outdoor farming. Predict the **result** before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain ranking, mechanism and generalization limits

- Five **observation**s cannot establish a stable ranking or causal **mechanism**; source and sampling assumptions remain important.

- Read the labelled picture: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the **result** establish that B is best for outdoor farming?

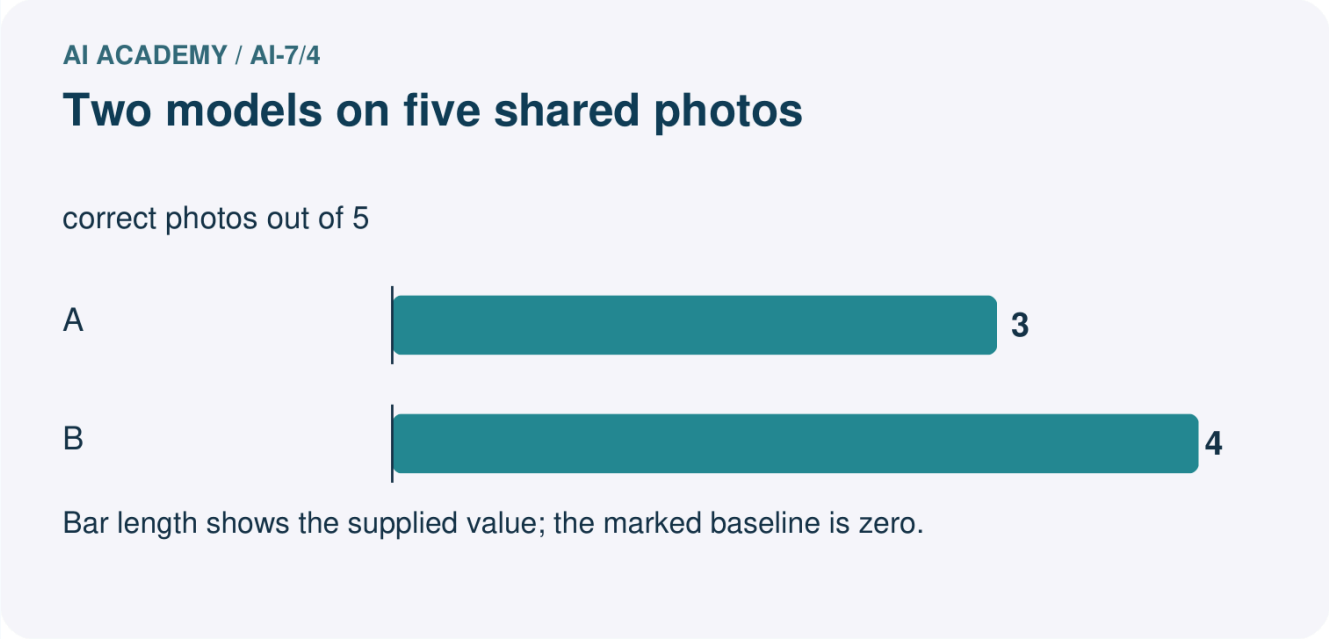
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the **result** before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the **result** and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Methods results and limitations

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/4

Methods results and limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the method

Same five photos, labels and scoring are used for both models.



2. State the result

A scores 60%; B scores 80% on this fixed set.



3. State the limit

The set supplies no outdoor-camera performance evidence.

CHANGE ONE THING

The conclusion says B is best for outdoor farming.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit role overlap and unsupported explanatory claims

> Differentiate “not studied” from “shown not to work”.

Core interpretation qualification and failure analysis

A **limitation** should constrain a claim rather than be hidden in unrelated prose.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised **method** and new **evidence**. Preserve unsuccessful cases so an improved **result** cannot hide the original problem.

Complete role-overlap and evidence-gap probes

Role-reading probe	Expected interpretation
Count with no denominator	Incomplete result statement; locate total and scoring
Same cases and labels	Method comparison condition, not the result itself
One reviewer assigned all labels	Method detail; agreement across reviewers not assessed
Single reviewer means labels must be wrong	Unsupported; limitation is not proof of error
No outdoor cases tested	Bounds interpretation; does not establish outdoor failure
A plausible mechanism explains a difference	Hypothesis, not an extra observation
One statement serves two roles	Name both with reasons rather than forcing a single label
Methods paragraph too brief to reproduce	Request missing details; do not invent settings
80% on fictional queries	No measured real-student learning effect follows

Guided repair and explanation

- A report omits this qualification: A **limitation** should constrain a claim rather than be hidden in unrelated prose. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise linking the sentence to the evidence it uses

> Give a reason for its role and a faithful alternative when needed.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example "**Method**s **result**s and **limitation**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Method**s **result**s and **limitation**s. Proposed explanation: “The worked **result** establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the **result** establish that B is best for outdoor farming?

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
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> | |

Reserve the selected cutoff/relevance/reviewer case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Complete a passage annotation and evidence-faithful rewrite

> Use classification to improve the argument rather than memorize labels alone.

Current twenty-query passage investigation

Statements: used 20 fictional queries; 16 passed; real student use was not studied. Show the intermediate steps and give the conclusion with its scope.

Complete annotation, arithmetic and rewrite protocol

Complete supplied-passage investigation	Expected written evidence
Copy the main three statements	20 fictional queries; 16 pass; real student use not studied
Annotate method/result/limitation	Name primary role and any scope overlap with reasons
Calculate the reported proportion	16/20 =80%; retain the supplied outcome and denominator
Write a faithful main summary	80% passed on these fictional queries; no classroom-learning evidence supplied
Classify separate picture facts	Same five photos/labels method; 3/5 vs4/5 result; no outdoor cases limitation
Audit the proposed outdoor conclusion	Replace with fixed-set result and missing-outdoor evidence
Add a mechanism hypothesis separately	Mark proposed explanation and a future test, not a newly observed cause
Record evidence status	Supplied-passage inspection/paper arithmetic; no live model or participant study performed

Expected evidence and unperformed-study limits

Use the fully supplied passages and counts for an annotation-and-rewrite investigation. Underline the procedure, circle the measured numerator/denominator and mark the boundary on interpretation. Explain any sentence with two roles. The main twenty-query passage and

separate five-photo comparison are different examples and must not share denominators.

No raw classifier outputs, reviewer-agreement data or real learner outcomes are supplied. This investigation needs no code or outside study; record your actual annotations and leave additional experiments unperformed.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

Try alone: cutoff, relevance fraction and one reviewer

> Classify the statements and explain overlapping roles.

Complete selected independent task

Statements: fixed cutoff 5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

First attempt and support record

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Statements: fixed cutoff 5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save method, reported results, limitations and hypotheses separately

> Connect each claim back to its supplied evidence.

Project task and artifact

Add a labelled **method**s **result**s and **limitation**s **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **method**s **result**s and **limitation**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and **limitation**. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	

Project evidence	Record
Failure, limit and human review	

Actual passage inspection and remaining evidence gaps

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the result without changing its outcome or scope

> A limitation makes the interpretation precise rather than inventing a new finding.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **method**s **result**s and **limitation**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Method	The procedure used to obtain observations
Result	A reported observation or calculated outcome
Limitation	A restriction on what results establish

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A **limitation** should constrain a claim rather than be hidden in unrelated prose. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the **result** alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Methods results and limitations — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Statements: used 20 fictional queries; 16 passed; real student use was not studied.	Classify each statement by role and link results to the procedure and denominator. A proposed mechanism is an explanation to test, not an extra observation.	First is method/scope; second result 16/20 = 80%; third limitation. No classroom effectiveness result is supplied.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its **result**. Cover the **result** first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.

By the end, I can	How I will show it
Explain the mechanism and terminology of methods results and limitations	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.
- Which labelled step matches this detail? The set supplies no outdoor-camera performance **evidence**.
- 1. State the **result**
- 2. State the limit
- 3. Identify the **method**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: State the limit The set supplies no outdoor-camera performance **evidence**.

Methods tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

Changed-case question and optional hint

- Changed-case practice: The conclusion says B is best for outdoor farming. Which conclusion is supported by the supplied facts?
- 1. Reject that extension without outdoor **evidence**.
- 2. This one example guarantees correct **result**s for every future **input**.
- 3. We can decide the **result** without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Reject that extension without outdoor **evidence**. The reported conditions do not include the proposed use population.

Repair and retain the record

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which **method**s and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

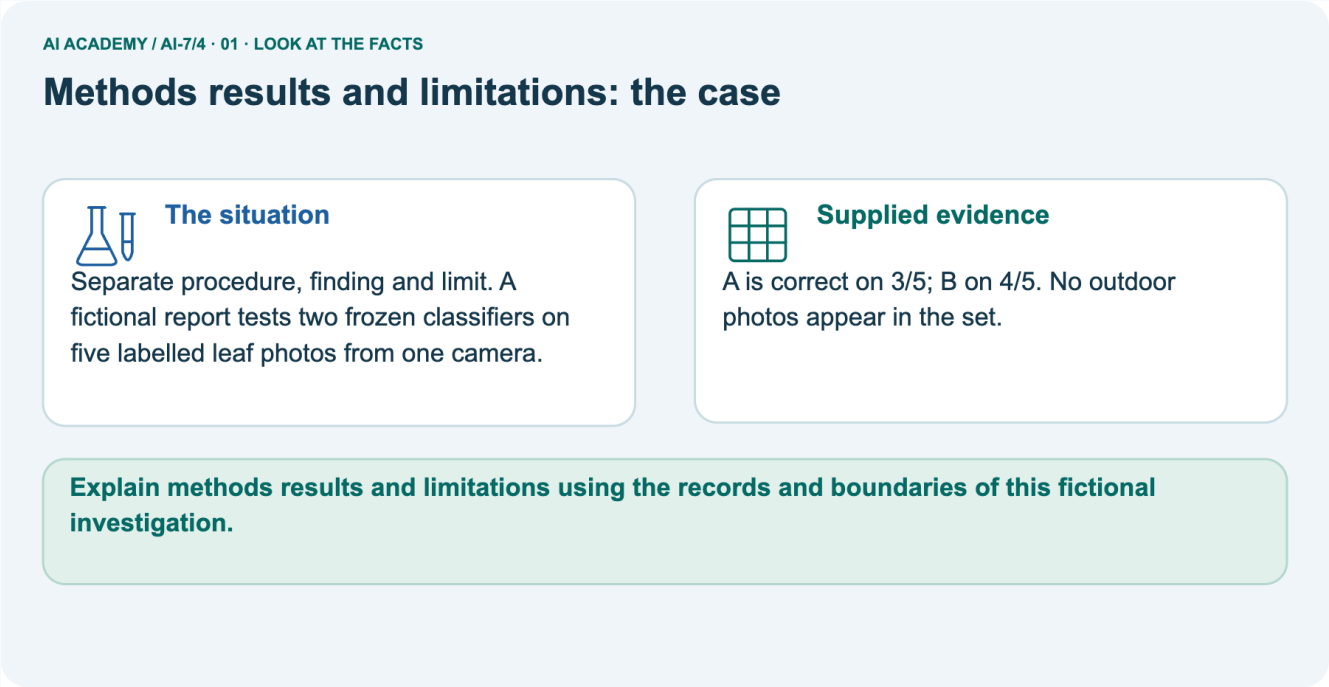
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Methods results and limitations: the case

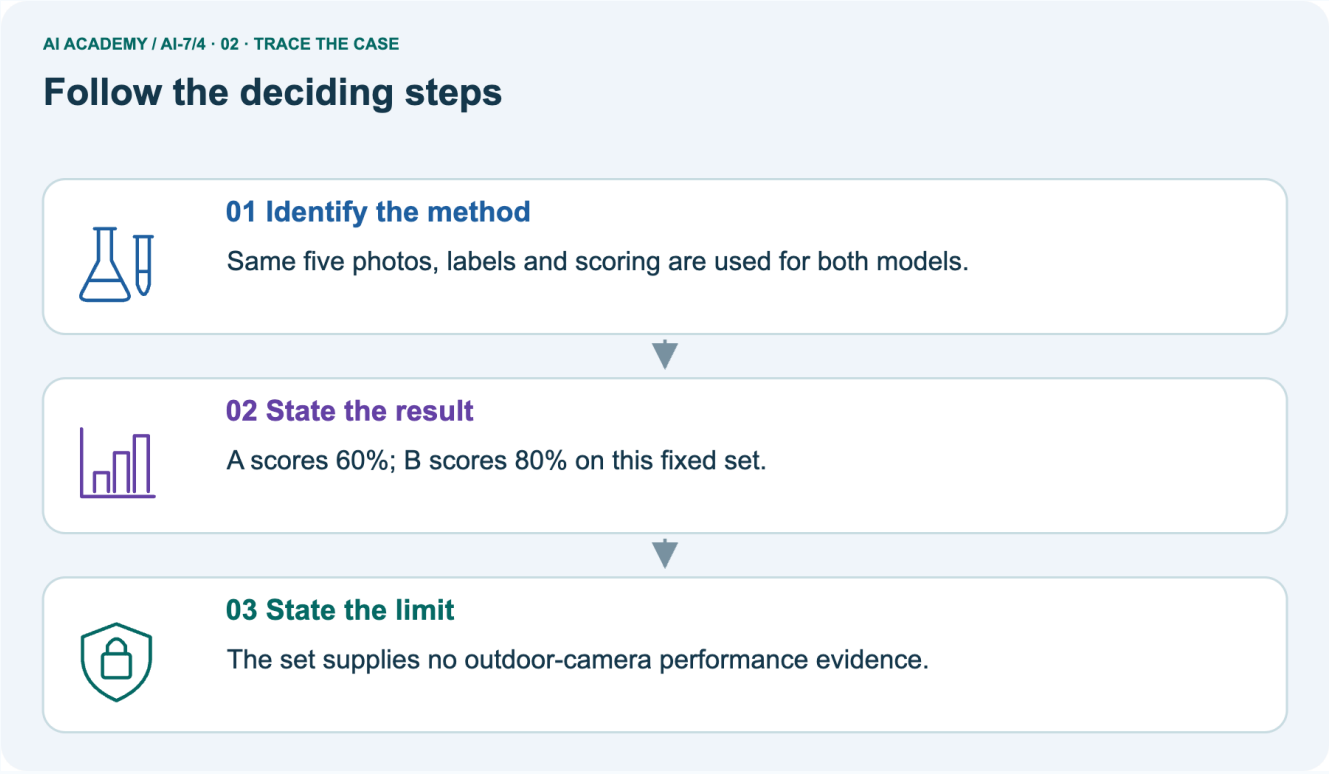


Follow the labelled picture with the matching explanation. Open the vector diagram

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera. Supplied facts: A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

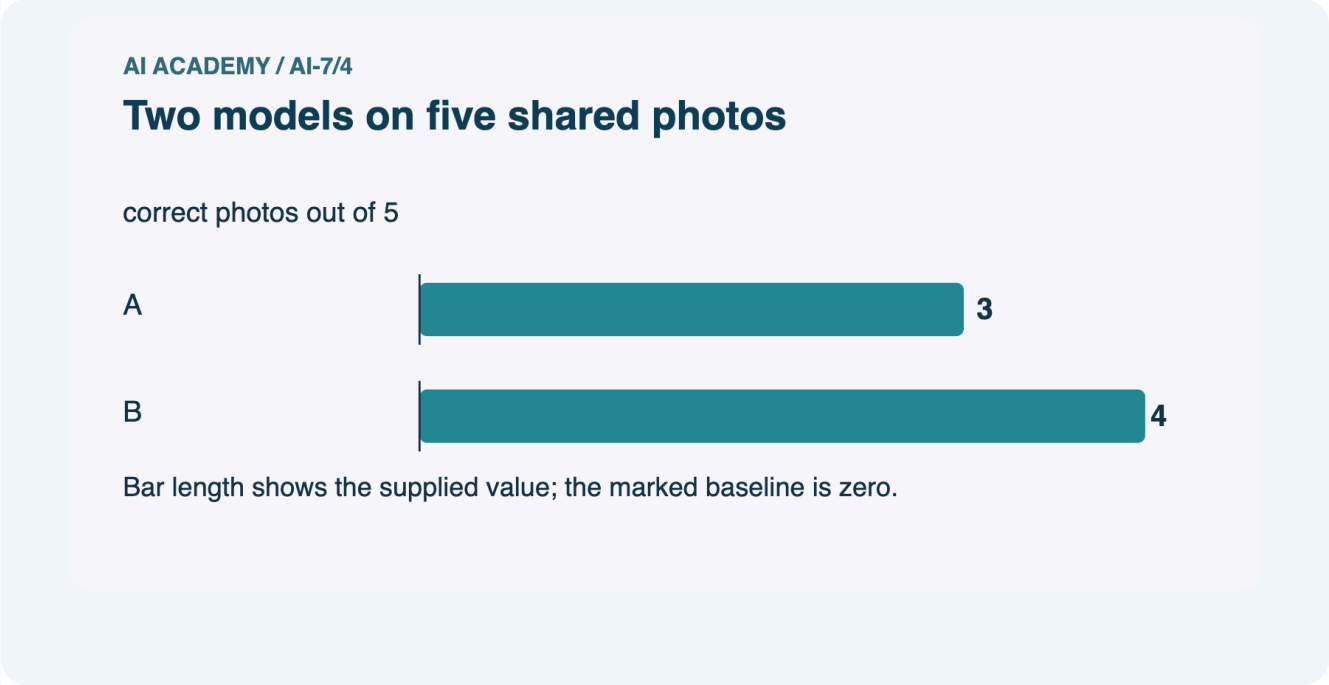


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the **method**: Same five photos, labels and scoring are used for both models. → State the **result**: A scores **60%**; B scores **80%** on this fixed set. → State the limit: The set supplies no outdoor-camera performance **evidence**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two models on five shared photos

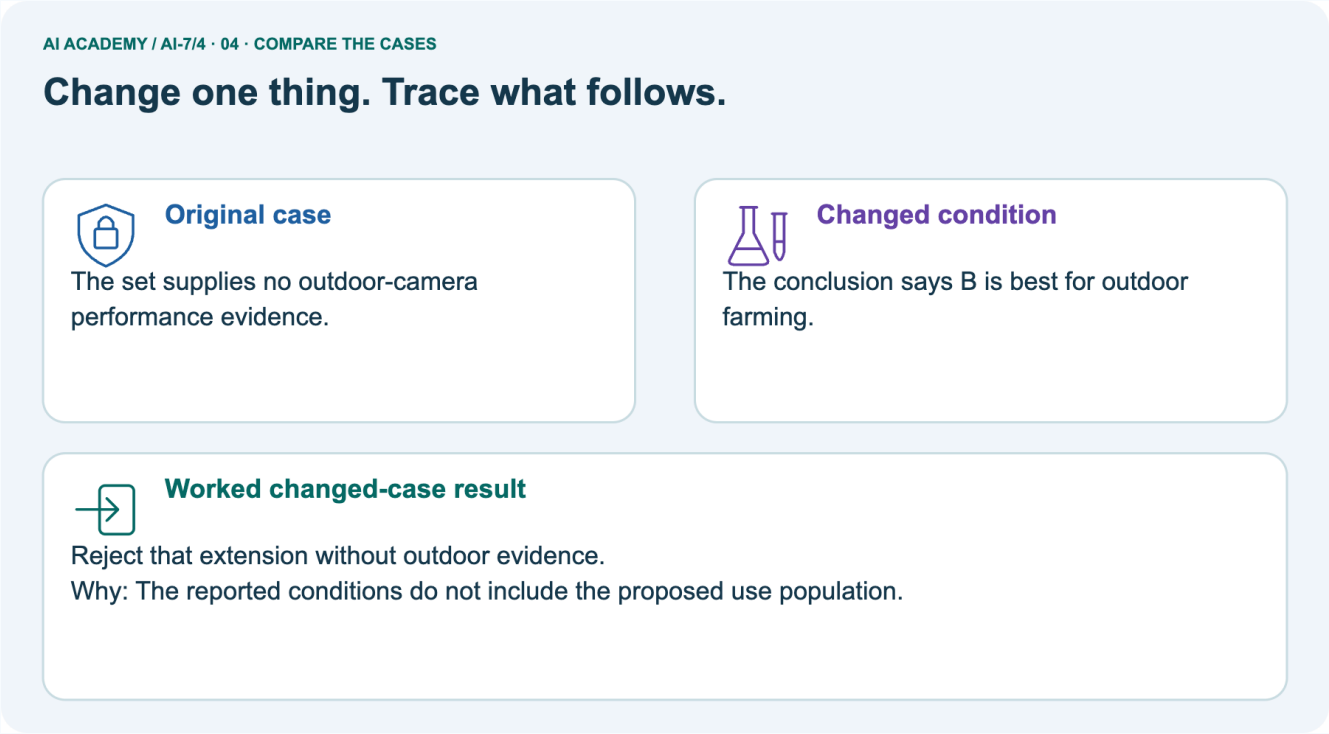


Follow the labelled picture with the matching explanation. Open the vector diagram

Two models on five shared photos. Units: correct photos out of 5. A: 3. B: 4.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



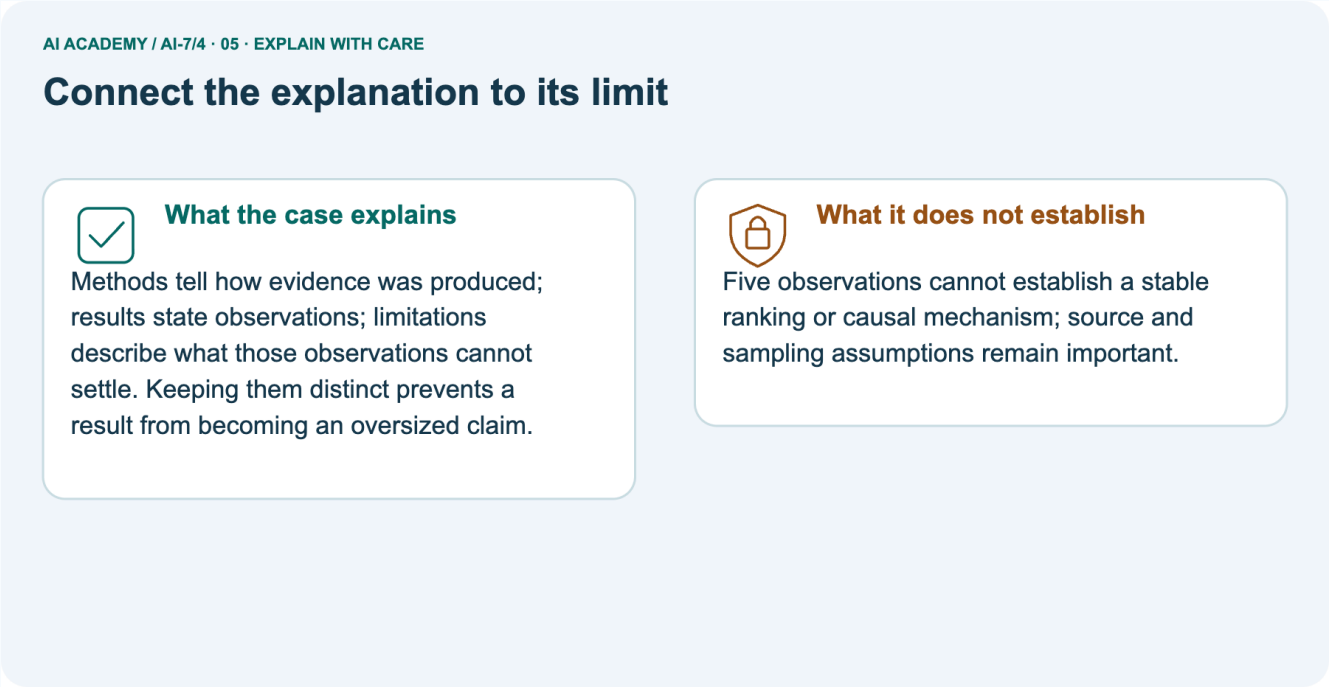
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The set supplies no outdoor-camera performance **evidence**. Changed condition: The conclusion says B is best for outdoor farming. Expected **result**: Reject that extension without outdoor **evidence**.

Reason: The reported conditions do not include the proposed use population.

Predict before checking the worked changed-case **result**. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



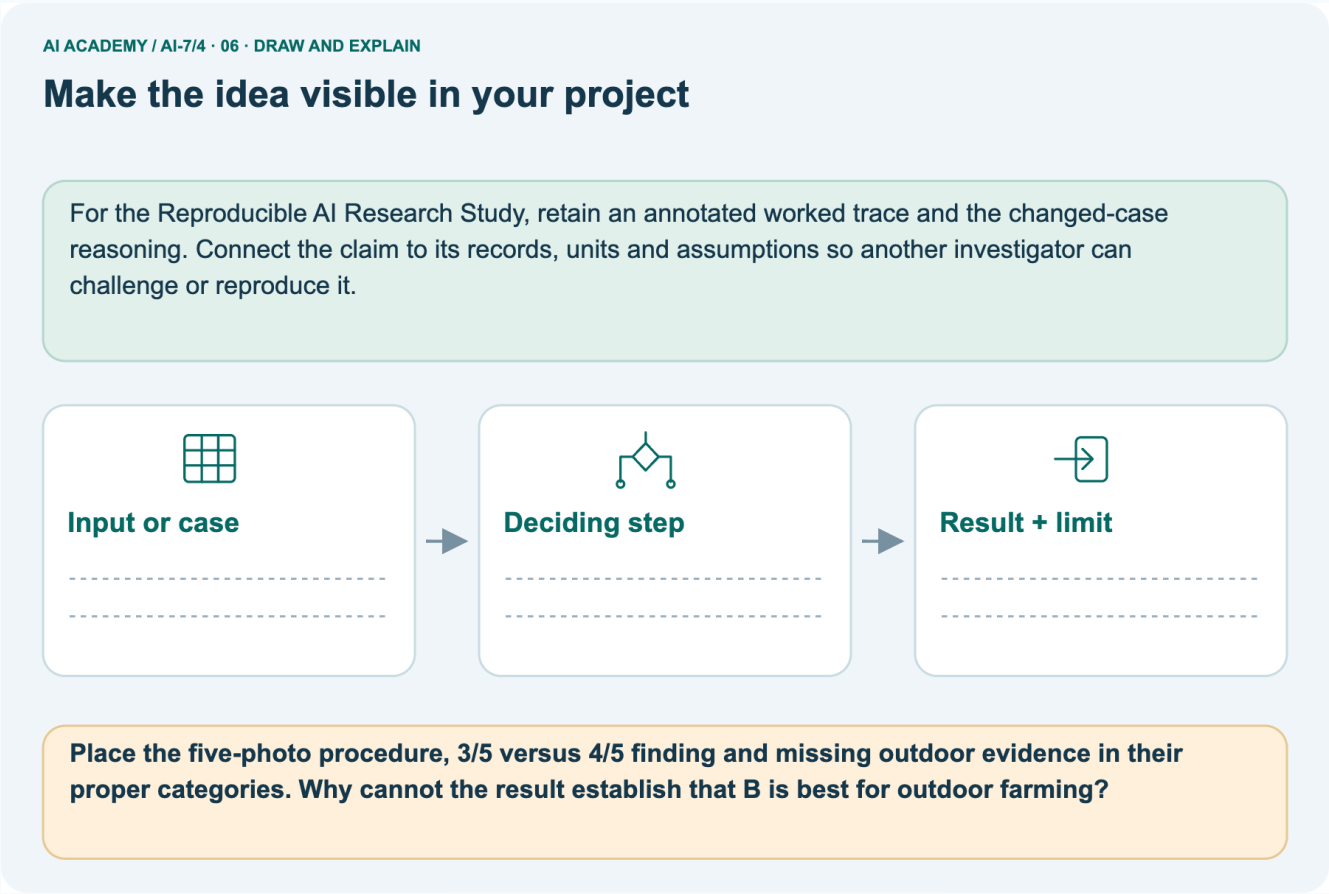
Follow the labelled picture with the matching explanation. Open the vector diagram

Methods tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

Limit: Five **observation**s cannot establish a stable ranking or causal **mechanism**; source and sampling assumptions remain important.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, **result** and limit. Supported question: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories.

Why cannot the **result** establish that B is best for outdoor farming?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Methods results and limitations: the case

02. Follow the deciding steps

03. Two models on five shared photos

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.